

# Useful Memory Has a Horizon

Temporal Validity, Roughness, and Structural Limits for  
Finite-Memory Tracking under Drift

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May 2026

## Abstract

In stationary settings, the law of large numbers makes memory an unambiguous statistical resource: more data improves estimation because the target distribution does not change over time. Under drift, this premise fails. Past evidence is generated by distributions that may no longer represent the present, so memory reduces variance while accumulating temporal misalignment. Once drift-induced misalignment dominates variance reduction, additional memory increases tracking error relative to the current distribution. Finite-memory tracking is therefore governed by a temporal-validity problem: how long can retained evidence continue to improve estimation of the present?

For drift path classes in which temporal misalignment grows as  $\zeta n^H$ , this variance-staleness trade-off induces a roughness-indexed useful-memory horizon  $n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}$  together with the corresponding minimax rates. The worst-case  $W_2$ -Lipschitz envelope corresponds to the  $H = 1$  member, recovering the cube-root horizon  $n^* \asymp (C_K/\zeta)^{2/3}$ . A Gaussian location lower bound shows that the same scaling defines a structural finite-memory floor.

This framework exposes a fundamental invalidity gap in non-stationary systems: retained evidence can lose temporal validity before a changepoint becomes statistically detectable. In finite samples, this structure appears through a finite optimum in horizon sweeps, roughness-dependent scaling across path classes, and measurable excess error when the operating horizon departs from the useful-memory scale.

**Keywords:** tracking under drift; useful memory; temporal validity; Wasserstein distance; Sinkhorn divergence; concept drift; finite-memory horizon.

**MSC 2020:** 93D05, 93D30, 68P99.

# 1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory therefore faces a basic question: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. More past data reduces sampling noise, but it also increases the age of the evidence used to represent the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The paper studies this tension as a horizon problem for temporal validity, and the main object is a roughness-indexed family of horizon laws.

Our main theorem package begins with a roughness-indexed trade-off between finite-sample variance and drift-induced staleness. In the worst-case  $W_2$ -Lipschitz envelope, that family specializes to the cube-root law and a structural finite-memory floor. A Gaussian location subclass provides a matching lower bound establishing the same structural finite-memory floor.

The same object extends beyond the worst-case Lipschitz regime. For deterministic Holder-type path classes, temporal roughness changes how staleness accumulates, and the horizon law becomes a family indexed by roughness. This makes the cube-root law the  $H = 1$  member of the family.

The practical conflict is between two clocks. One clock measures the age at which retained evidence stops being valid for the present. The other measures when accumulated data provide enough statistical evidence for a changepoint alarm. Those clocks need not move together. The central conceptual distinction is therefore simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained memory is already stale for the present.

The contributions are fivefold:

- It formalizes useful memory under drift as a variance-staleness trade-off in Wasserstein geometry.
- It proves a worst-case finite-memory horizon law under local  $W_2$ -Lipschitz drift.
- It shows a Gaussian-location lower bound establishing a structural tracking floor in the critical-window regime.
- It extends the same analysis to Holder-type path classes and their roughness-indexed horizon laws.
- It identifies detector-silent staleness and measures the invalidity gap between temporal validity and changepoint evidence.

The empirical evidence follows that logic in stages. It first recovers the predicted  $U$ -curve and the finite useful-memory optimum. It then shows the gap between temporal validity and changepoint evidence, the regime-dependent scaling across roughness classes, and the nonlinear cost of horizon misalignment. Real-stream evidence, when kept, serves only as weak confirmation of the same structural picture.

The theoretical results are stated in  $W_2$ ; the Sinkhorn divergence is the computational measurement layer used in the synthetic experiments. The window size

result of [Hinder et al. \[2023\]](#) motivates the broader issue, but the contribution here is more specific: memory has a finite temporal validity horizon under worst-case drift, and that horizon law extends naturally to Holder-type path classes.

## 2 Temporal Validity Under Drift

Under drift, memory is both a statistical resource and a temporal liability. Retaining more data reduces finite-sample noise, but it also increases the age of the evidence used to represent the present. The structural question is therefore not whether memory helps in the abstract, but how long retained evidence can remain temporally valid before staleness overwhelms the remaining variance gain.

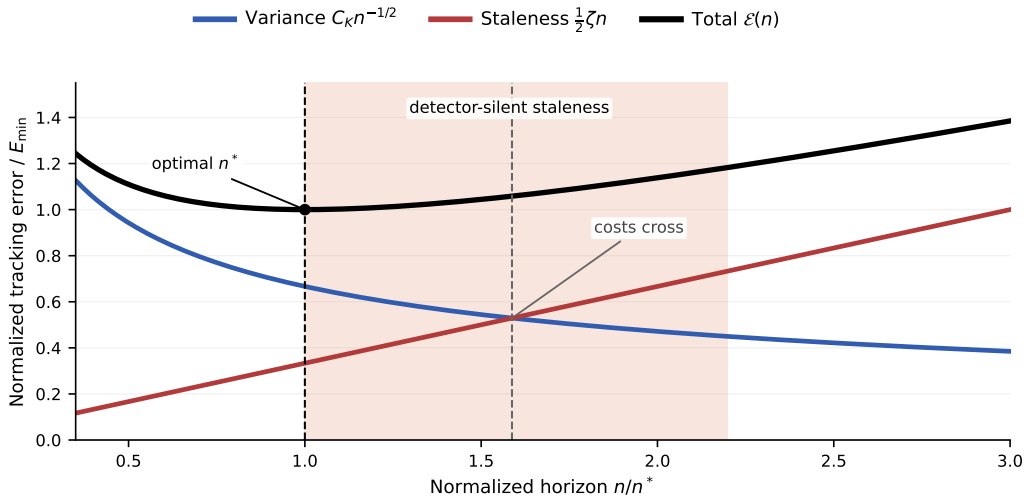


Figure 1: Two clocks of drift in normalized theorem units. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the useful-memory horizon  $n^*$ . The shaded region marks detector-silent staleness: temporal invalidity can arrive before changepoint evidence.

For a finite-memory estimator with lag weights  $w_0, \dots, w_{n-1}$ , the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j w_j.$$

Its operational staleness at time  $t$  is the weighted misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j W_2(P_{t-j}^*, P_t^*).$$

Under a local  $W_2$ -Lipschitz drift envelope with rate  $\zeta_t$ ,

$$S_t(w) \leq \zeta_t A(w).$$

For a uniform window,  $A(w) = (n-1)/2$ , so temporal age and window length are the same object up to constants. This is why horizon choice becomes the central statistical question.

## 2.1 Roughness-Indexed Horizon Family

The family of horizon laws comes first. Temporal roughness controls how staleness accumulates with lag, so different path classes induce different useful-memory scales.

**Proposition 2.1** (Roughness-indexed horizon law). *Let a deterministic Holder-type path class have roughness index  $H \in (0, 1]$  and staleness growth*

$$\text{Staleness}_H(n; \zeta) \asymp \zeta n^H.$$

*If the finite-sample term remains of order  $C_K n^{-1/2}$ , then the tracking error bound*

$$\mathcal{E}_H(n) \leq C_K n^{-1/2} + c_H \zeta n^H$$

*is minimized at*

$$n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}, \quad \mathcal{E}_{H,\min} \asymp C_K^{2H/(1+2H)} \zeta^{1/(1+2H)}.$$

*In particular,  $H = 1$  recovers the cube-root law.*

*Proof.* Balancing the decreasing statistical term against the increasing staleness term gives

$$C_K n^{-1/2} \asymp \zeta n^H,$$

which yields

$$n^{H+1/2} \asymp C_K/\zeta, \quad n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}.$$

Substituting this scale back into either term gives the stated minimum error.  $\square$

*Remark 2.2* (Family interpretation). The point is not that the cube-root law is wrong. The point is that it is the  $H = 1$  member of a broader roughness-indexed family. As  $H$  decreases, staleness accumulates more slowly with lag, so the useful-memory horizon moves accordingly.

## 2.2 Worst-Case Lipschitz Envelope

The worst-case Lipschitz regime is the linear-staleness envelope inside that family. It isolates the sharpest age penalty permitted by the local path bound.

**Theorem 2.3** (Lag-inclusive tracking error bound). *Let  $X_{t-j} \sim P_{t-j}^*$  and let*

$$\widehat{P}_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{X_{t-j}}$$

*be the uniform-window empirical estimator. Assume the target path is locally  $W_2$ -Lipschitz with rate  $\zeta$ , so that*

$$W_2(P_{t-j}^*, P_t^*) \leq \zeta j \quad \text{for all } 0 \leq j < n,$$

*and assume the finite-sample term obeys*

$$\mathbb{E} \left[ W_2 \left( \widehat{P}_t^{(n)}, \bar{P}_t^{(n)} \right) \right] \leq C_K n^{-1/2},$$

*where  $\bar{P}_t^{(n)} := n^{-1} \sum_{j=0}^{n-1} P_{t-j}^*$ . Then the expected tracking error satisfies*

$$\mathcal{E}(n) \leq C_K n^{-1/2} + \frac{1}{2} \zeta n,$$

*where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.*

*Proof.* Apply the triangle inequality:

$$W_2\left(\widehat{P}_t^{(n)}, P_t^*\right) \leq W_2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + W_2\left(\bar{P}_t^{(n)}, P_t^*\right).$$

The first term is controlled by the assumed finite-sample bound. For the second term, let  $\pi_j$  be an optimal coupling between  $P_{t-j}^*$  and  $P_t^*$ . The averaged coupling  $\bar{\pi} := n^{-1} \sum_{j=0}^{n-1} \pi_j$  couples  $\bar{P}_t^{(n)}$  with  $P_t^*$ , so

$$W_2\left(\bar{P}_t^{(n)}, P_t^*\right) \leq \frac{1}{n} \sum_{j=0}^{n-1} W_2(P_{t-j}^*, P_t^*) \leq \frac{1}{n} \sum_{j=0}^{n-1} \zeta j \leq \frac{1}{2} \zeta n.$$

Combining the two bounds gives the claim.  $\square$

**Lemma 2.4** (Finite-sample Wasserstein term). *Let  $Q$  be a probability measure on  $\mathbb{R}^{d_{\text{eff}}}$  with support contained in a ball of radius  $R$  and finite second moment. Let  $X_1, \dots, X_n$  be i.i.d. samples from  $Q$ , and let  $\hat{Q}_n := n^{-1} \sum_{i=1}^n \delta_{X_i}$ . Then:*

- If  $d_{\text{eff}} < 4$ , there exists a constant  $C_{d_{\text{eff}}, R}$  such that

$$\mathbb{E} W_2(\hat{Q}_n, Q) \leq C_{d_{\text{eff}}, R} n^{-1/2}.$$

- If  $d_{\text{eff}} = 4$ , the same bound holds with an additional  $(\log n)^{1/2}$  factor.
- If  $d_{\text{eff}} > 4$ , the raw Wasserstein rate is slower than  $n^{-1/2}$ , so the constant-only form above should not be used.

In that high-dimensional regime, the statistical fluctuations do not follow the  $n^{-1/2}$  carrier exponent, so the resulting horizon scaling lies outside the displayed cube-root regime. If the geometry is computed with debiased Sinkhorn divergence  $S_\varepsilon$  on a bounded domain of diameter  $D$ , then for fixed  $\varepsilon > 0$  the empirical fluctuations remain  $O(n^{-1/2})$  with a constant  $L(D, \varepsilon)$  that is dimension-free at fixed  $\varepsilon$  and increases as  $\varepsilon \downarrow 0$ .

A short numerical sanity check for this regime appears in Appendix A; it is not used as a theorem, but it matches the intended low-dimensional interpretation of the finite-sample term.

The two terms in theorem 2.3 pull in opposite directions. Increasing  $n$  reduces variance but raises the age of the retained evidence. That tension produces the worst-case Lipschitz horizon law.

**Proposition 2.5** (Optimal horizon in the Lipschitz case). *Under theorem 2.3, the error  $\mathcal{E}(n)$  is minimized at*

$$n^* = \left( \frac{C_K}{\zeta} \right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}.$$

*Proof.* Differentiate the upper envelope  $C_K n^{-1/2} + \frac{1}{2} \zeta n$  and solve

$$-\frac{1}{2} C_K n^{-3/2} + \frac{1}{2} \zeta = 0.$$

This gives  $n^* = (C_K/\zeta)^{2/3}$ . Substituting back yields the displayed minimum.  $\square$

*Remark 2.6* (Scope of the horizon law). The cube-root law follows when the finite-sample term behaves as  $\mathbb{E} W_2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-1/2}$ . More generally, if the finite-sample term admits scaling

$$\mathbb{E} W_2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-a} \quad \text{for some } a > 0,$$

then balancing it against the linear staleness cost  $\frac{1}{2}\zeta n$  yields

$$n^*(a) = \left( \frac{2a C_K}{\zeta} \right)^{\frac{1}{a+1}}, \quad \mathcal{E}_{\min}(a) = \frac{a+1}{(2a)^{a/(a+1)}} C_K^{1/(a+1)} \zeta^{a/(a+1)}.$$

The main text uses only the  $a = 1/2$  regime. The displayed cube-root law is therefore the worst-case linear-staleness member of the broader family.

*Remark 2.7* (Measurement layer). The theorem is stated in  $W_2$ , so  $\varepsilon$  does not enter the drift law itself. In the synthetic experiments, the observable geometry is often a debiased Sinkhorn divergence with fixed  $\varepsilon$ , which keeps the same  $n^{-1/2}$  carrier exponent while changing constants. The computational layer is therefore separate from the structural horizon law.

**Corollary 2.8** (EWMA analogue). *For exponential weights  $w_j = \alpha(1 - \alpha)^j$ , the effective lag is  $\tau_\alpha = (1 - \alpha)/\alpha$  and the effective sample size scales as  $n_{\text{eff}} \asymp 1/\alpha$ . Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale. This is the exponentially weighted analogue of the  $H = 1$  case.*

## 2.3 A Structural Finite-Memory Floor

The upper law is not merely the optimum of one estimator. The same scale appears in a lower-bound witness, which makes the useful-memory horizon structural for the stated class.

**Proposition 2.9** (Finite-memory floor via a Gaussian location witness). *Let  $\mathcal{P}_{\zeta, m}$  be the class of length- $m$  distribution paths such that:*

1. *each path is  $W_2$ -Lipschitz with rate at most  $\zeta$ ;*
2. *the lower-bound construction is carried out on the Gaussian location subclass  $X_t \sim \mathcal{N}(\mu_t, \sigma^2 I_d)$  with  $\|\mu_t - \mu_s\|_2 \leq \zeta|t - s|$ ;*
3. *the estimator at time  $m$  only uses the last  $m$  samples.*

*Measure risk by the endpoint  $W_2$  error, which coincides with Euclidean endpoint error on the Gaussian location subclass. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:*

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E} d(\hat{\mathcal{P}}_m, \mathcal{P}^*) \gtrsim \max\{C_K m^{-1/2}, c \sigma^{2/3} \zeta^{1/3}\}.$$

*Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural finite-memory floor on this class.*

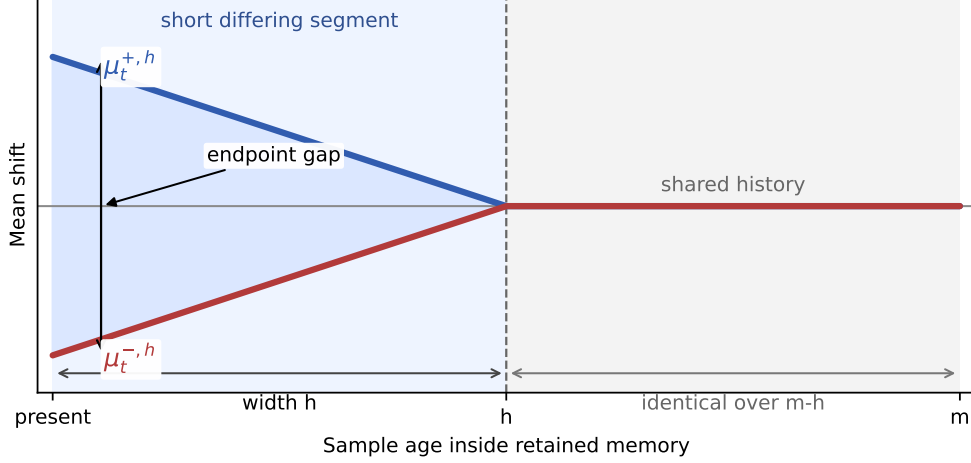


Figure 2: Lower-bound witness on the Gaussian location subclass. The two paths share most of the retained history and differ only on a short recent segment of width  $h$ , yet they remain separated at the present. That unresolved recent segment is enough to create a structural finite-memory floor.

**Proof.** Write  $R_m^* := \inf_{\delta_m} R_m(\delta_m)$ . We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass  $\mu_s \equiv \theta$ , which is contained in the admissible drift class. For the two hypotheses  $\theta_{\pm} = \pm\sigma/(2\sqrt{m})$ , the induced product-Gaussian laws have Kullback–Leibler divergence

$$\text{KL}(P_+, P_-) = \frac{1}{2\sigma^2} \sum_{j=1}^m (\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left( \frac{\sigma}{\sqrt{m}} \right)^2 = \frac{1}{2}.$$

For any estimator  $\delta_m$ , the plug-in test  $\psi = \mathbf{1}\{\delta_m \geq 0\}$  must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least  $\sigma/(2\sqrt{m})$ . By Le Cam’s two-point lemma and Pinsker’s inequality, the average testing error is at least

$$\frac{1 - \text{TV}(P_+, P_-)}{2} \geq \frac{1 - \sqrt{\text{KL}(P_+, P_-)/2}}{2} \geq \frac{1}{4}.$$

Hence

$$R_m^* \geq \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is  $c_0 = 1/8$ .

The second obstruction is local drift. Fix any  $1 \leq h \leq m$  and define

$$\beta_h := \min\left\{\zeta, \frac{\sigma}{4h^{3/2}}\right\}, \quad \mu_{t-j}^{\pm,h} := \pm\beta_h(h-j)_+, \quad 0 \leq j \leq m-1,$$

where  $(u)_+ := \max\{u, 0\}$ . Each path belongs to the admissible class because its slope magnitude is at most  $\beta_h \leq \zeta$ . The endpoint gap is

$$|\mu_t^{+,h} - \mu_t^{-,h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\text{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2} \sum_{j=0}^{m-1} (2\beta_h(h-j)_+)^2 \leq \frac{2\beta_h^2}{\sigma^2} \sum_{r=1}^h r^2 \leq \frac{2\beta_h^2}{\sigma^2} h^3 \leq \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold  $\beta_h h$ , and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1 - \text{TV}(P_{+,h}, P_{-,h})}{2} \geq \frac{1 - \sqrt{\text{KL}(P_{+,h}, P_{-,h})/2}}{2} \geq \frac{3}{8}.$$

Therefore

$$R_m^* \geq \frac{3}{8} \beta_h h \geq \frac{3}{32} \min\{\zeta h, \sigma h^{-1/2}\},$$

which is again a fully explicit numerical lower bound. Since  $h$  was arbitrary,

$$R_m^* \geq \frac{3}{32} \sup_{1 \leq h \leq m} \min\{\zeta h, \sigma h^{-1/2}\}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume  $1 \leq h_* := (\sigma/\zeta)^{2/3} \leq m$ . Choose an integer  $\bar{h} \in [h_*/2, h_*]$ . Then

$$\zeta \bar{h} \geq \frac{1}{2} \zeta h_* = \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and

$$\sigma \bar{h}^{-1/2} \geq \sigma h_*^{-1/2} = \sigma^{2/3} \zeta^{1/3}.$$

Therefore

$$\min\{\sigma \bar{h}^{-1/2}, \zeta \bar{h}\} \geq \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \geq \frac{3}{64} \sigma^{2/3} \zeta^{1/3}.$$

So one admissible explicit constant in the critical-window regime is  $c = 3/64$ .

*Remark 2.10* (Scope of the lower bound). The lower bound is subclass-based. It uses a Gaussian location witness whose mean path is itself  $W_2$ -Lipschitz, and the distance in the lower bound is the endpoint  $W_2$  error, which becomes Euclidean on that subclass. That is enough to make the result minimax for any larger class containing the same witness, including the ambient  $W_2$ -Lipschitz path class. The proposition therefore does not claim that every path is Gaussian; it claims that the worst case already appears inside the Gaussian mean-shift subproblem.

### 3 Empirical Evidence

The experiments make the theory visible in finite samples. They do not compare named interventions. They read observed tracking behavior against the useful-memory scale implied by the regime under study. Throughout, the oracle reference is the horizon predicted by the same variance–staleness law that drives the theory section.

Four signatures matter. First, horizon sweeps should exhibit a finite optimum. Second, that optimum should scale differently across roughness classes. Third, the operating horizon can lose temporal validity before a detector accumulates enough evidence to signal change. Fourth, excess error should increase once the operating horizon departs materially from the useful-memory scale.

### 3.1 Structural U-Curve

The Gaussian sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the optimum lies at an interior horizon rather than at either boundary.

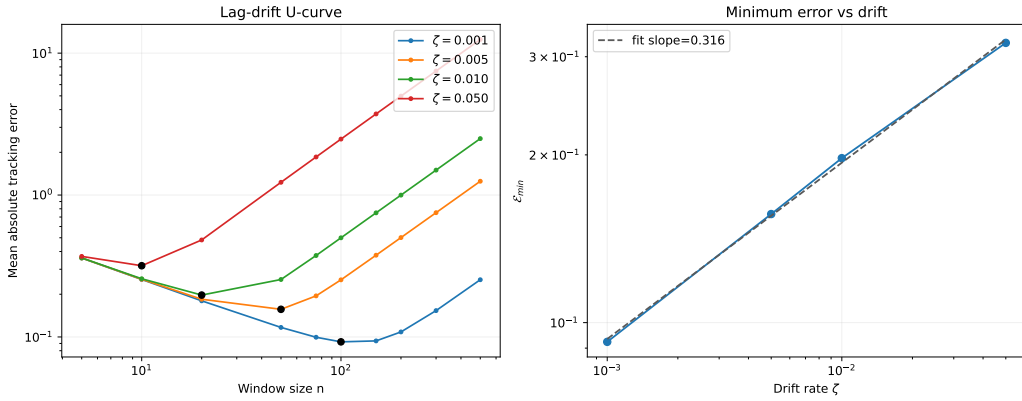


Figure 3: Structural U-curve for finite-memory tracking. Left: tracking error as a function of horizon for several drift strengths. Right: the minimum achievable error decreases as drift weakens. The optimum remains finite across regimes, which makes the useful-memory horizon visible before any path-class refinement.

This is the basic finite-memory signature. Under stationarity, more memory would continue to reduce variance. Under drift, the sweep bends upward once temporal misalignment overtakes the remaining variance gain.

### 3.2 Regime-Dependent Scaling Across Roughness Classes

The next experiment varies the roughness index directly. For each Holder-type regime, the benchmark samples a finite-sample surrogate whose error obeys the same variance–staleness decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

then measures the empirically optimal horizon across a sweep of  $\zeta$  values.

The point of this figure is structural rather than cosmetic. If the cube-root law were the whole story, the right panel would collapse to a single slope. It does not. The empirical optima separate by roughness class, and the fitted slopes track the predicted family in the exported roughness-scaling summaries.

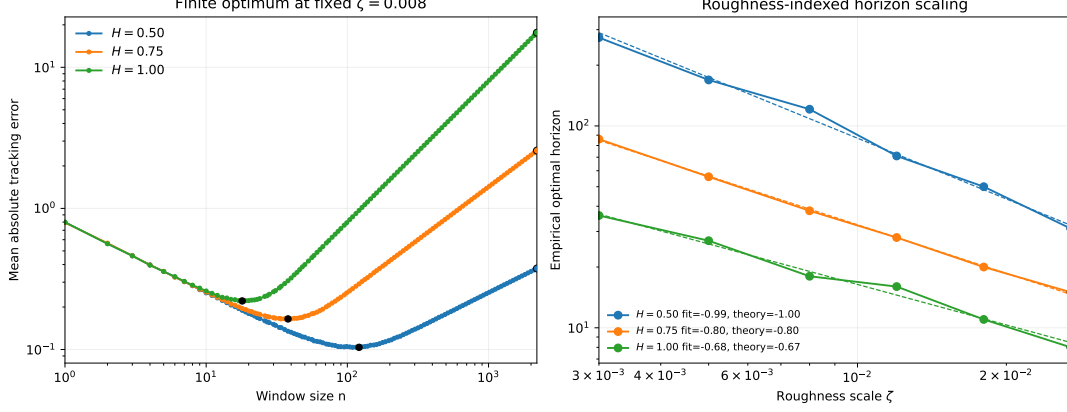


Figure 4: Roughness-dependent horizon scaling. Left: each roughness class still produces a finite optimum, but the location of that optimum shifts with  $H$ . Right: the empirical optimal horizon follows a distinct log-log scaling law in each class. The  $H = 1$  curve recovers the worst-case Lipschitz member, while smaller  $H$  values produce different exponents inside the same family.

### 3.3 Detector-Silent Staleness

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The useful-memory horizon  $n_t^*$  contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{inv}} := t_{\text{detect}} - t_{\text{valid}},$$

where  $t_{\text{valid}}$  is the first time the operating horizon persistently exceeds the useful-memory scale, and  $t_{\text{detect}}$  is the first detector alarm that follows.

Across detector settings, the measured gap remains positive in every run of the default benchmark. The detector-threshold ablation shows mean gaps of roughly  $5 \times 10^2$  steps over the tested settings. This is the observable form of detector-silent staleness: evidence can become obsolete while standard detection remains silent.

### 3.4 Horizon Misalignment Cost

The final question is how sharply performance degrades when the operating horizon misses the useful-memory scale. Using the same roughness-indexed surrogate, fig. 6 plots excess error against the relative horizon gap  $|\log(n/n^*)|$ .

This is the finite-sample reason the horizon itself matters. The optimum is not only finite; it is operationally relevant. Errors remain modest in a narrow band around the useful-memory scale, then increase on either side as variance or staleness takes over.

### 3.5 What the Evidence Shows

Taken together, the experiments reveal four features that match the theory. First, finite-memory tracking exhibits a structural U-curve rather than a boundary optimum. Second, the useful-memory scale is roughness dependent, with the worst-case

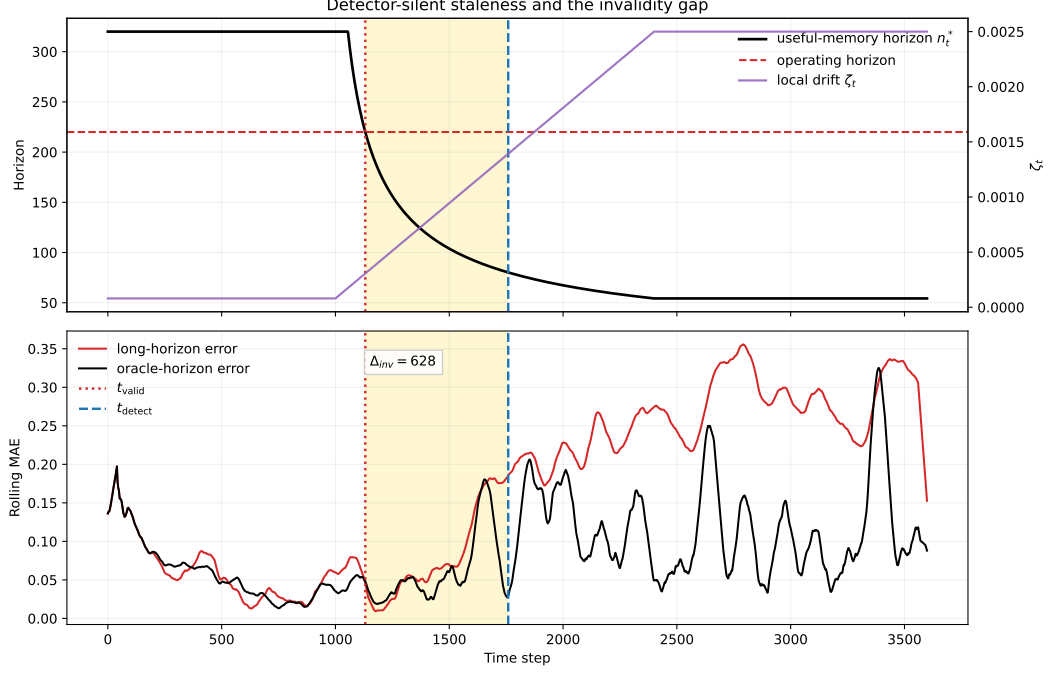


Figure 5: Detector-silent staleness. As drift intensifies, the useful-memory horizon contracts until the operating horizon becomes stale at  $t_{\text{valid}}$ ; the detector responds only later at  $t_{\text{detect}}$ , leaving a strictly positive invalidity gap. In the lower panel, stale memory already imposes excess tracking error before the alarm appears.

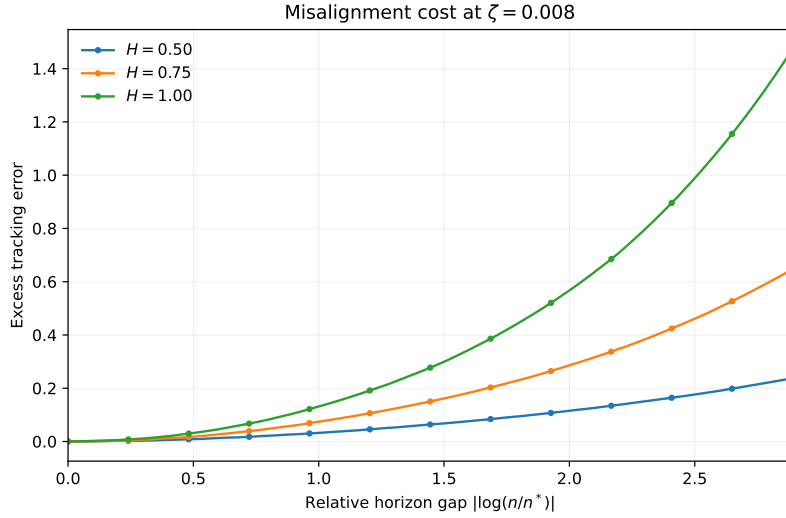


Figure 6: Cost of horizon misalignment. Excess tracking error is smallest near the useful-memory scale and rises as the operating horizon moves away from it. Being too short is noisy, being too long is stale, and the penalty steepens at class-dependent rates once the operating horizon leaves the low-cost band around  $n^*$ .

Lipschitz cube-root law appearing as the  $H = 1$  member of a wider family. Third, temporal validity can fail before changepoint evidence appears, which produces a measurable invalidity gap. Fourth, missing the useful-memory scale carries a visible finite-sample cost even when the operating horizon is only moderately misaligned.

## 4 Related Work

**Nonstationary optimization and dynamic regret.** Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

**Adaptive windowing and drift detection.** Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. The invalidity gap makes that distinction measurable rather than rhetorical.

**Distribution tracking under transport geometry.** Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

**Wasserstein nonstationarity and robust optimization.** Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

**Freshness and Age of Information.** The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory has a roughness-indexed temporal horizon, the worst-case cube-root law is the  $H = 1$  member of that family, and temporal validity can fail before change becomes statistically detectable.

## 5 Limitations and Scope

**Path-class scope.** The main upper law is stated for deterministic path classes in which staleness can be controlled as a function of lag. The worst-case Lipschitz result is the  $H = 1$  member of that family, while other roughness classes induce different horizon exponents.

**Lower-bound scope.** The structural floor is proved through a Gaussian location witness embedded in the ambient  $W_2$ -Lipschitz class. That is enough for a subclass-based minimax lower bound, but it does not characterize the full extremal geometry of the entire class. The exponent is structural; the exact hardest path and the sharp constant are left open.

**Finite-sample carrier regime.** The clean cube-root law uses the regime in which the finite-sample term behaves like  $C_K n^{-1/2}$ . That is the relevant low-dimensional Wasserstein regime and also the regime induced by fixed- $\varepsilon$  Sinkhorn surrogates on bounded domains. Outside that carrier exponent, the same balancing argument gives a different horizon law, as noted in remark 2.6.

**Detectability versus validity.** The invalidity gap is an empirical consequence of the structural theory, not the primary theorem object. The paper shows that temporal validity can fail before a detector fires, but it does not claim a universal detection-delay theorem across all detector classes.

**What remains open.** Several questions remain deliberately open: sharp constants in the horizon laws, matching lower bounds for broader Holder classes, online identification of the relevant roughness class, and transfer from deterministic path-class statements to stronger stochastic-process formulations. Those are natural next steps, but they are not prerequisites for the main structural claim.

## 6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting variance–staleness trade-off makes useful memory a finite-horizon object rather than an unlimited statistical resource.

The analysis shows that this horizon is roughness indexed. For path classes in which temporal misalignment grows like  $\zeta n^H$ , the useful-memory scale is

$$n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The cube-root law is therefore not the whole theory. It is the worst-case Lipschitz,  $H = 1$ , member of a broader family. The Gaussian location lower bound shows that the same scale is structural rather than an artifact of one particular estimator.

The empirical section makes the same structure visible in finite samples. It reveals a structural U-curve, roughness-dependent scaling across path classes, a strictly positive invalidity gap between temporal validity and changepoint evidence, and measurable cost when the operating horizon departs from the useful-memory scale.

The paper therefore supports a simple conclusion. Useful memory has a finite horizon determined by how temporal roughness accumulates staleness, and retained evidence can become invalid before change becomes statistically detectable.

What remains open is equally clear: sharper constants, fuller lower-bound characterizations beyond the Gaussian witness, and online identification of the relevant roughness class. Those questions refine the theory; they do not alter its main object.

## Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

## A Finite-Sample Geometry Check

For the finite-sample term in lemma 2.4, we estimate empirical  $W_2$  between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log-log slope in  $n$ . We also check the Sinkhorn proxy in one dimension at a few fixed values of  $\varepsilon$ .

Table 1: Empirical scaling check for the finite-sample term. The bounded-support  $W_2$  experiment is close to the expected  $n^{-1/2}$  rate in low dimension, and the slope degrades as intrinsic dimension increases. The Sinkhorn proxy is reported at fixed  $\varepsilon$ , where the measured slope and constant both vary with regularization.

| Setting                       | Estimated slope | Comment                |
|-------------------------------|-----------------|------------------------|
| $W_2, d = 1$                  | -0.48           | close to $-1/2$        |
| $W_2, d = 2$                  | -0.41           | slower decay           |
| $W_2, d = 4$                  | -0.25           | slower decay           |
| $W_2, d = 5$                  | -0.24           | slower decay           |
| Sinkhorn $\varepsilon = 0.50$ | -0.21           | proxy constant changes |
| Sinkhorn $\varepsilon = 0.10$ | -0.53           | proxy constant changes |
| Sinkhorn $\varepsilon = 0.05$ | -0.67           | proxy constant changes |

The numerical check is only meant to support the modeling choice behind lemma 2.4. In low dimension, the raw Wasserstein term is compatible with an  $n^{-1/2}$  fluctuation scale, whereas higher-dimensional raw Wasserstein decays more slowly.

For fixed  $\varepsilon$ , the Sinkhorn proxy acts as a measurement layer whose calibration constant changes with regularization and ambient geometry even when the horizon argument keeps the same formal shape.

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